Vacation Rental Marketplace

PRODUCTION ARCHITECTURE

Enterprise Scalable System Design • Multi-Region Edge • Microservices • Event-Driven Storage

FRONTEND React 19 + TypeScript + Vercel Edge

BACKEND Stateless Kubernetes / Node.js & Go Microservices

SCALE 100M+ Monthly Visits • 99.99% SLA

1. CLIENT & EDGE DELIVERY TIER

CLIENT LAYER

🖥️ Desktop & Web Client

SPA

High-fidelity Airbnb clone frontend with dynamic listing views, Photo Tour, Lightbox modal, interactive booking widget, and responsive layouts.

`React 19` `TypeScript` `Vite 6` `Vanilla CSS` `URL Param State` `Keyboard A11y`

🌐 Edge Network & CDN

Vercel Edge

Global Anycast Edge network handling TLS termination, asset caching, DDoS mitigation, and edge header rewrites.

`Global PoPs` `Brotli / Gzip` `HTTP/3 (QUIC)` `WebP Optimizer` `WAF & DDoS Shield`

🚀 CI/CD & Deployment

Automated

GitHub Actions automated pipeline with strict linting, type-checking, bundle optimization, and zero-downtime atomic deployments.

`GitHub Actions` `Vercel CLI` `Preview Deployments` `Production Rollouts` `Secret Vault`

⚡ Client Performance & Cache

`Cache-Control Immutable` `Service Worker Prefetch` `IntersectionObserver` `Sub-100ms FCP`

2. API GATEWAY — Kong / Envoy Traffic Director

`JWT Auth & RBAC` `Token Bucket Rate Limiter` `Circuit Breaker` `SSL Offload` `gRPC Web Transcoding`

3. CORE MICROSERVICES TIER (HORIZONTALLY SCALED KUBERNETES CLUSTER)

STATELESS COMPUTE

🔐 Auth & Identity

OAuth 2.0 / OIDC, JWT Token issuance, 2FA verification, Session management.

`Node.js` `Redis Session`

🏠 Listing & Property

Property metadata, 43-photo gallery configs, sleeping arrangements, amenities.

`Go` `PostgreSQL`

🔍 Search & Discovery

Geospatial bounding-box queries, price/amenity filtering, availability ranking.

`Java` `OpenSearch`

📅 Booking & Reserve

Real-time calendar locks, double-booking prevention, reservation lifecycle.

`Go` `Redlock`

💳 Payment & Escrow

Stripe integration, guest checkout, host payout automation, refund ledger.

`Go` `PCI-DSS Vault`

⭐ Reviews & Ratings

Category scoring (cleanliness, location, etc.), distribution math, Guest Favourite.

`Node.js` `PostgreSQL`

👤 User & Host Profile

Host verification, Superhost status, co-host permissions, messaging.

`Node.js` `PostgreSQL`

🔔 Notifications

Real-time WebSockets, mobile push alerts, transactional email and SMS dispatch.

`Rust` `WebSockets`

4. Asynchronous Event Bus (Apache Kafka / RabbitMQ)

`listing.updated` `booking.confirmed` `payment.succeeded` `review.submitted`

5. DATA PERSISTENCE & STORAGE LAYER

HIGH AVAILABILITY

🎮 Primary DB (Master)

Amazon Aurora PostgreSQL Multi-AZ with synchronous replication for ACID write consistency.

`Write Master`

📖 Read Replicas

Horizontally scaled Aurora Read Replicas serving high-volume listing reads with connection pooling.

`PgBouncer` `Auto-scale`

⚡ Redis Cache Cluster

Distributed cache for fast listing details, rate limits, active user sessions, and calendar states.

`Sub-millisecond`

🔎 OpenSearch Cluster

Inverted index supporting geo-queries, autocomplete, faceted filtering, and price aggregations.

`Geo-spatial Index`

📦 S3 / R2 Media Storage

High-durability blob storage for 43+ property gallery images, WebP assets, and PITR backups.

`11 9's Durability` `CDN Origin`

6. SCALING, SECURITY & OPS

INFRASTRUCTURE

📈 High-Scale Strategy

Engineered for extreme traffic spikes with multi-layer caching, stateless container replicas, and async queue processing.

`Horizontal Pod Autoscaler (HPA)` `Multi-Zone Redundancy` `Distributed Redlock` `Database Partitioning` `Edge Asset Caching`

📊 Observability & Telemetry

Datadog

Full-stack distributed tracing, application performance monitoring (APM), and proactive anomaly detection.

`OpenTelemetry Traces` `Prometheus Metrics` `Grafana Dashboards` `ELK Log Aggregation` `PagerDuty Alerts`

🛡️ Security & Compliance

Zero Trust

Defense-in-depth security with automated secret rotation, mutual TLS (mTLS), and strict network segmentation.

`AWS Secrets Manager` `mTLS Inter-service` `PCI-DSS Compliant` `GDPR Data Isolation` `Continuous SAST / DAST`

🔄 Backup & Disaster Recovery

`Continuous Aurora PITR` `Cross-Region Replication` `RTO < 15 mins` `RPO < 1 min`

Legend: HTTPS / Client Traffic · gRPC Microservice RPC · Async Kafka Events · Database Read/Write

`Edge Caching` `Stateless Scale` `Distributed Locking` `Read Replicas` `99.99% Availability`